# Zephyr Sensor Fusion with MicroBlaze V on Kria KR260 and OpenAMP

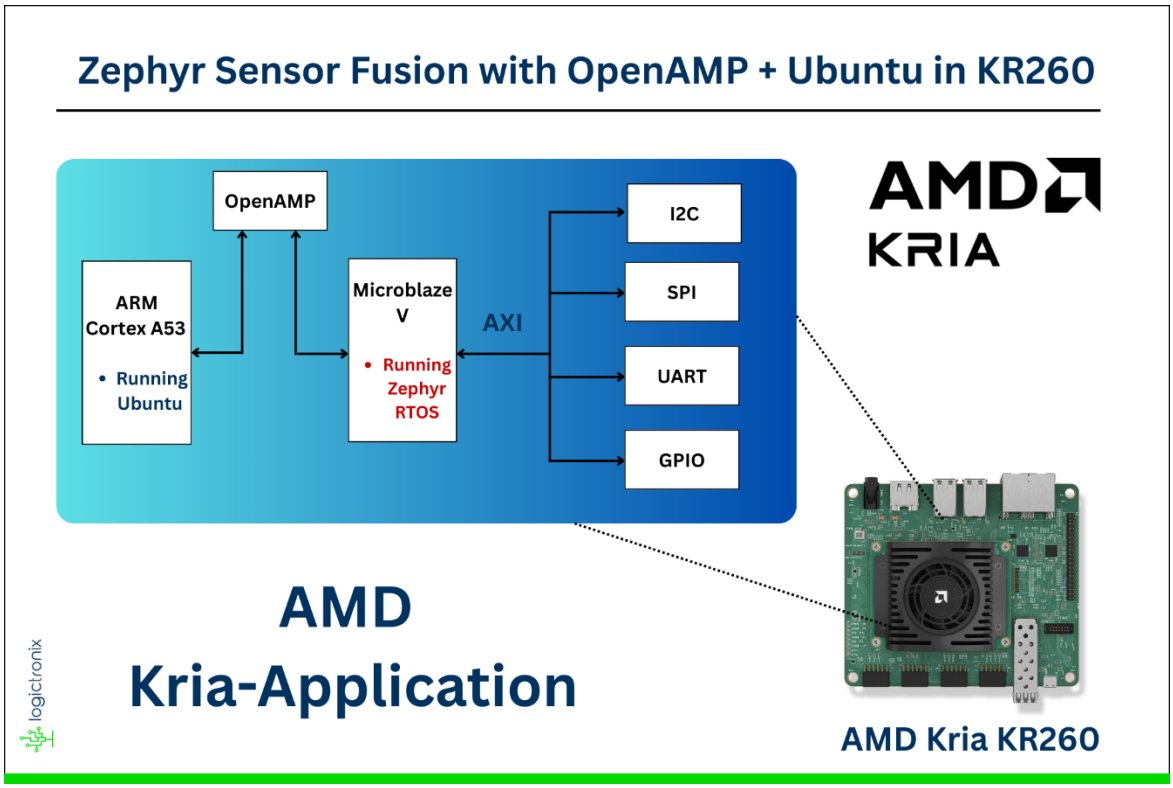
*Getting Started Guide*

## A. Overview

This Kria Application is on Running Zephyr RTOS application for Sensor Fusion on Microblaze V along with OpenAMP on top of Ubuntu/Linux on the KR260 Board's MPSoC PS (A53) core.

This Kria application showcase the sensor fusion and multi-OS integration with the Kria Robotics Starter Kit (KR260). We are utilizing Digilent Pmod Sensors for Temperature and Accelerometer along with Pmod based UART and GPIO.

**Architecture of the Kria App:**



## B. Preparing the KR260 and Sensor Setup

1. **Necessary Hardware - KR260 and Pmod Accessories**
   a. KR260 Board , power adapter , 16+ GB of SD Card – Purchase Link from AMD
   b. Pmod ACL, accelerometer – Digilent Link
   c. Pmod TMP2, temperature sensor– Digilent Link
   d. Pmod GPIO 8 bit LED – Digilent Link
   e. Pmod UART – Digilent Link

2. **Preparing the KR260 Board with Ubuntu 22.04 LTS image**
   a. Follow the "**Booting Kria Starter Kit Linux on KR260**" steps from here - **ubuntu_22_04 [booting linux on KR260]**.
   b. Where you will perform:
      - Download the image
      - Burn SD card using Balena etcher
      - Update the boot firmware at KR260
      - And boot the KR260 with Ubuntu 22.04LTS.

3. **Connecting the Pmod Sensors with KR260 Robotics Starter Kit:**
   Below are the details for connection of Pmod sensors and modules at KR260.
   **a. Pmod Acl - Accelerometer**
      - Connect to Pmod1.

   **b. Pmod TMP2 – Temperature sensor**
      - Connect to Pmod3.

   **c. Pmod GPIO – 8 bit LED**
      - Connect to Pmod4.

   **d. Pmod UART**
      - Connect Pmod-UART to Pmod2 – lower as shown in picture below.
      - You can follow following VIVADO Constraint section for connection.
        ```
        # UART_txd PMOD #2 pin 4 (connect rx of the uart device to pin 4)
        # UART_rxd PMOD #2 pin 6 (connect tx of the uart device to pin 6)
        ```

   **e. LAN connection for Git Clone**
      - Connect the LAN connection (via router for internet) – LAN connection is needed to clone this GitHub repo to KR260 board image and also accessing it via SSH.

4. **Connect the Pmod-UART and KR260 UART on the board with PC.**
    a. If you have two "***USB-A to micro-B Cable***" then connect one to J4 (JTAG/UART) port at KR260 and another to Digilent Pmod-UART.
        - The J4 (JTAG/UART) connection is primarily required to determine the IP address assigned to the KR260 once the boot process is complete. After obtaining the IP address, it can be used to access the KR260 via SSH.
        - ***[Optional]*** If you only have one USB-A to micro-B cable, first connect it to the J4 (JTAG/UART) port of the KR260. After identifying the KR260's IP address, you can disconnect it from J4 and reconnect it to the Pmod-UART port.
    b. You can use Mobaxterm/Teraterm in ***Windows*** or Gtkterm in ***Linux***.
    c. Connect the baudrate of **115200** for both of the UARTs

5. **KR260 Pmod sensors and connection:**

6. **Now boot the KR260 board.**
    a. After booting , you will start getting boot log from Ubuntu 22.04LTS into your PC/computer.
    b. After boot completes, you can enter the username – **ubuntu** and password – which you have setup in Step- B.2.
    c. ***[Optional]*** You can also identify and node down the KR260's IP address by running the ***ifconfig*** command at the terminal of KR260. Once you have identified the IP address, you can disconnect the USB-A to micro-B cable from J4 and reconnect it to the Pmod-UART port.

## C. Clone Github Repository

Clone this github repository at the KR260-Ubuntu image by running git clone command:

```
git clone https://github.com/LogicTronixInc/MicroBlaze-V-Zephyr-Kria-OpenAMP.git
```

## D. Download and install the packages

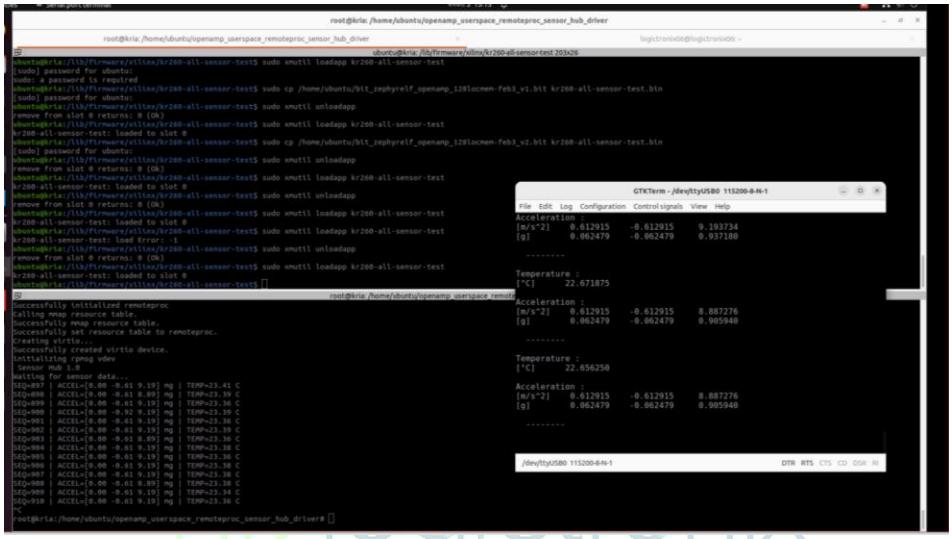
- For installing device tree, kernel, firmware, libmetal library , openamp library and to make host code file, please follow this readme document and steps from here - Kria-Ubuntu/README.md

## E. Run the Application

Please follow the same readme document for running the application at KR260 Ubuntu (host app) and KR260 Zephyr (sensor fusion app) frimware - [Kria-Ubuntu/README.md]

**Here are also the steps brief:**

1. You can connect the KR260 via SSH (for multi terminal access) as like here:



2. Unload the default Kria app and load the sensor fusion kria app. This process also loads the Bitstream as well as Zephyr application (***zephyr.elf***) inside (merged-in) it.
3. Now, run the OpenAMP host app on the Kria-Ubuntu terminal.
4. You should now be able to see the output on the UART terminal, including temperature and accelerometer (X, Y, Z) values reflecting changes in acceleration, as shown in the image above. In addition to viewing the sensor data in Ubuntu, you can also observe the data on the Zephyr UART side. The temperature values are also displayed on the Pmod-GPIO LED.

**Congratulations for successful run and completion of this Kria-App.**

**For any queries, please raise the GitHub issue or mail us at info@logictronix.com.**

## Revision History

| S.N | Revision Number | Date of Revision | Revision Details | Author |
|---|---|---|---|---|
| 1. | v1.0 | February 11th 2026 | Initial Revision | Krishna Gaihre and Sanam Shakya |
| 2. | v1.2 | March 25th 2026 | Updated the setup and run flow | Krishna Gaihre and Sanam Shakya |